TDA reproduction: results by dataset

Karmanov et al., CVPR 2024 · Bocconi HPC cluster · 30 October 2026

Table 3: cross-domain (completed)

Dataset	RN50 ours	RN50 paper	Δ	ViT-B/16 ours	ViT-B/16 paper	Δ
Caltech101	89.53	89.70	−0.17	94.04	94.24	−0.20
DTD	43.56	43.74	−0.18	45.04	47.40	−2.36
EuroSAT	41.07	42.11	−1.04	63.06	58.00	+5.06
FGVC Aircraft	17.22	17.61	−0.39	25.02	23.91	+1.11
Food101	77.80	77.75	+0.05	86.18	86.14	+0.04
Flowers102	67.88	68.74	−0.86	71.46	71.42	+0.04
Oxford Pets	85.85	86.18	−0.33	89.64	88.63	+1.01
UCF101	63.73	64.18	−0.45	71.05	70.66	+0.39
Stanford Cars	*missing (official links no longer work)*			*missing*		
SUN397	*missing (37 GB, exceeds available storage)*			*missing*		

Top-1 accuracy (%). Δ = ours − paper. Green: within 1 point · orange: about 1 point · red: more than 1 point.

Table 1: out-of-distribution (ViT completed, RN50 running)

Dataset	RN50	ViT-B/16 CLIP	ViT-B/16 paper	ViT-B/16 ours	Δ
ImageNet-A	*running*	59.72	60.11	60.30	+0.19
ImageNet-V2	*running*	64.40	64.67	64.45	−0.22
ImageNet-R	*running*	80.30	80.24	80.41	+0.17
ImageNet-Sketch	*running*	49.52	50.54	50.93	+0.39
OOD average (4)			**63.89**	**64.02**	**+0.13**
ImageNet	*missing*	—	69.51	*missing*	

Δ = ours − paper. The four variants are separate datasets that share ImageNet's 1000 classes, so the paper's OOD average can be compared even without ImageNet.

> **Comment.** Our re-implementation produces exactly the same results as the authors' official code on every dataset (difference 0.00). With **RN50** the reproduction succeeds: 7 of 8 datasets are within 1 point of the paper, with EuroSAT at −1.04. With **ViT-B/16** 4 of 8 datasets match; the other 4 deviate by 1–5 points, in both directions. Since the official code gives these same numbers, the likely causes are the single random ordering of test images used in the paper and the published hyperparameters; we will check this by repeating the experiments over multiple orderings. In Table 1 with **ViT-B/16** all four datasets are within 0.4 points of the paper (OOD average +0.13); RN50 is still running. ImageNet, Stanford Cars and SUN397 are missing, and therefore so are the ablations, which use ImageNet only.